

**Table SX.** Model summaries for intercept-only models for species within cat prey range. Model estimates  $\pm$  95% CIs and test statistics are reported along with sample sizes and residual unexplained heterogeneity ( $I^2$ ), decomposed by hierarchical model levels (article and observation). The effect size used is given in the heading of each model (SMD=standardized mean difference or Hedges' g; Zr=correlation coefficient; lnOR=log odds ratio).

| Term                                                 | Estimate                  | Test statistics           | Sample size                                                          | Residual heterogeneity                                             |
|------------------------------------------------------|---------------------------|---------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Abundance with/without cats   SMD</b>             |                           |                           |                                                                      |                                                                    |
| overall                                              | -0.4 $\pm$ [-1.14, 0.34]  | $t_8 = -1.25, p = 0.25$   | $N_{articles} = 9$<br>$N_{species} = 12$<br>$N_{observations} = 14$  | $I^2_{total} = 54.8$<br>$I^2_{article} = 54.8$<br>$I^2_{obs} = 0$  |
| <b>Abundance before-after eradication   SMD</b>      |                           |                           |                                                                      |                                                                    |
| overall                                              | -0.91 $\pm$ [-8.73, 6.91] | $t_1 = -1.48, p = 0.38$   | $N_{articles} = 2$<br>$N_{species} = 2$<br>$N_{observations} = 3$    | $I^2_{total} = 38$<br>$I^2_{article} = 38$<br>$I^2_{obs} = 0$      |
| <b>Abundance on islands with/without cats   lnOR</b> |                           |                           |                                                                      |                                                                    |
| overall                                              | 0.03 $\pm$ [-1.7, 1.76]   | $t_5 = 0.04, p = 0.97$    | $N_{articles} = 6$<br>$N_{species} = 10$<br>$N_{observations} = 10$  | $I^2_{total} = 42.1$<br>$I^2_{article} = 42.1$<br>$I^2_{obs} = 0$  |
| <b>Abundance spatial association   Zr</b>            |                           |                           |                                                                      |                                                                    |
| overall                                              | 0.1 $\pm$ [-1.25, 1.45]   | $t_3 = 0.24, p = 0.83$    | $N_{articles} = 4$<br>$N_{species} = 2$<br>$N_{observations} = 4$    | $I^2_{total} = 67.3$<br>$I^2_{article} = 33.6$<br>$I^2_{obs} = 34$ |
| <b>Abundance temporal association   Zr</b>           |                           |                           |                                                                      |                                                                    |
| overall                                              | 0.1 $\pm$ [-0.37, 0.57]   | $t_{10} = 0.47, p = 0.65$ | $N_{articles} = 11$<br>$N_{species} = 12$<br>$N_{observations} = 18$ | $I^2_{total} = 85$<br>$I^2_{article} = 56.6$<br>$I^2_{obs} = 28$   |
| <b>Reproduction with/without cats   lnOR</b>         |                           |                           |                                                                      |                                                                    |
| overall                                              | -0.39 $\pm$ [-3.52, 2.75] | $t_2 = -0.53, p = 0.65$   | $N_{articles} = 3$<br>$N_{species} = 3$<br>$N_{observations} = 5$    | $I^2_{total} = 65.5$<br>$I^2_{article} = 28.2$<br>$I^2_{obs} = 37$ |
| <b>Reproduction before-after eradication   SMD</b>   |                           |                           |                                                                      |                                                                    |
| overall                                              | 0.14 $\pm$ [-5.7, 5.99]   | $t_1 = 0.31, p = 0.81$    | $N_{articles} = 2$<br>$N_{species} = 3$<br>$N_{observations} = 3$    | $I^2_{total} = 54.4$<br>$I^2_{article} = 54.4$<br>$I^2_{obs} = 0$  |
| <b>Reproduction temporal association   Zr</b>        |                           |                           |                                                                      |                                                                    |

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| Term    | Estimate                | Test statistics        | Sample size                                                       | Residual heterogeneity                                        |
|---------|-------------------------|------------------------|-------------------------------------------------------------------|---------------------------------------------------------------|
| overall | 0.09 $\pm$ [-0.62, 0.8] | $t_3 = 0.41, p = 0.71$ | $N_{articles} = 1$<br>$N_{species} = 1$<br>$N_{observations} = 4$ | $I^2_{total} = 0$<br>$I^2_{article} = NA$<br>$I^2_{obs} = NA$ |